Joint Planning of Distribution Systems and EV Charging Infrastructure Using a GIS-Based Spatial Analysis Framework

Mario A. Mejia, *Member, IEEE*, Leonardo H. Macedo, *Member, IEEE*, Gregorio Muñoz-Delgado, *Senior Member, IEEE*, Javier Contreras, *Fellow, IEEE*, and John F. Franco, *Senior Member, IEEE*

***Abstract*— The availability of public charging infrastructure is essential for accelerating the widespread adoption of electric vehicles (EVs). However, because EVs consume significant amounts of electricity, it is essential to coordinate the installation of this infrastructure with distribution system planning to prevent network failures. This article introduces an innovative spatial analysis framework based on Geographic Information Systems (GIS) designed to enable the integrated planning of distribution systems and public electric vehicle charging infrastructure (EVCI). The proposed framework consists of three modules. The first module estimates the spatial locations of potential EV users based on socioeconomic data and calculates the charging demand profiles for these vehicles. The second module examines the spatial relationships between distribution system nodes, road infrastructure, popular venues, and urban land use restrictions to determine which nodes are best suited for EVCI installation. The third module optimizes the location and size of the EVCI by integrating the results of the previous two modules with the distribution system's physical, operational, and topological characteristics. Additionally, the third module determines which assets need to be installed in the distribution system to support the increased load. The framework's performance was tested using geospatial data from a Brazilian city. The findings suggest that planning EVCI without considering the spatial features of the study area can lead to impractical or unfeasible solutions.**

***Keywords*— Active distribution system planning, electric vehicle charging infrastructure, geographic information systems, scenario-based stochastic programming, spatial analysis.**

I. Introduction

THE shift to electric vehicles (EVs) is gaining traction worldwide as a promising solution for lowering greenhouse gas emissions and improving urban air quality [1]. Aside from environmental benefits, this transition aims to reduce reliance on fossil fuels, promote green mobility, and reshape the energy landscape toward sustainability [2]. However, a robust public charging infrastructure is fundamental to encourage EV adoption and expedite the move towards an eco-friendlier transportation system. This infrastructure must be designed to be accessible across the road network, offering a safe and efficient charging experience to instill confidence in potential EV users [1].

Planning electric vehicle charging infrastructure (EVCI) entails sizing and strategically locating chargers to meet the EV charging demands within urban areas [2]. This process considers factors such as population density, EV users' travel patterns, and the availability of electrical energy to charge EVs [3]. However, this task is hindered by several challenges. These challenges include the lack of uniform patterns in EV adoption, making it difficult to appropriately size and strategically locate EVCI [4]. Additionally, limited capacity within the distribution system in certain regions may necessitate costly upgrades to accommodate the EV charging demands [5]. Other obstacles include the availability of physical space for EVCI installation, easy access to EVCI via road networks, and ensuring the proximity of EVCI to popular venues such as shopping malls, sets of offices, hospitals, universities, and residential areas [6].

Moreover, inadequate EVCI planning can significantly impact distribution system operations. To begin, a sudden increase in energy demand in regions with limited installed capacity may overload transformers and distribution lines, resulting in power outages [7]. Furthermore, an excessive concentration of EVCI in specific locations may cause grid congestion and voltage drops, reducing service quality [8]. Furthermore, poor EVCI planning may result in imbalances in distribution system load, necessitating costly infrastructure upgrades to meet rising demand [9]. Likewise, an improperly located EVCI may fail to fully utilize available renewable energy sources, complicating the transition to a more sustainable distribution network [10]. Therefore, EVCI installation must be coordinated with distribution system planning (DSP) to avoid power system operational breakdowns.

A. Literature review

B. Contributions

This paper proposes an innovative GIS-based spatial analysis framework designed to enable the integrated planning of distribution systems and EVCI. The proposed framework consists of three modules. The first module estimates the spatial locations of potential EV users. The second module determines which distribution system nodes are best suited for EVCI installation. The third module optimizes the location and size of the EVCI.

The previous methods determine the proper location for EVCI installation based on the overlap of the nodes of both networks. However, there may be land use restrictions that prevent the EVCI from being built in the places where the nodes of both networks intersect [6]. Thus, in this paper, in

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M. A. Mejia and J. F. Franco are with the Department of Electrical Engineering, São Paulo State University (UNESP), School of Engineering, Ilha Solteira, SP, 15385-000, Brazil (e-mails: mario.andres@unesp.br; fredy.franco@unesp.br).

L. H. Macedo is with the Department of Engineering, São Paulo State University (UNESP), School of Engineering and Sciences, Rosana, SP, 19274-000, Brazil (e-mail: leohfmp@ieee.org).

G. Muñoz-Delgado and J. Contreras are with the Escuela Técnica Superior de Ingenieros Industriales, Universidad de Castilla-La Mancha, 13071, Ciudad Real, Spain (e-mails: Gregorio.Munoz@uclm.es; Javier.Contreras@uclm.es).

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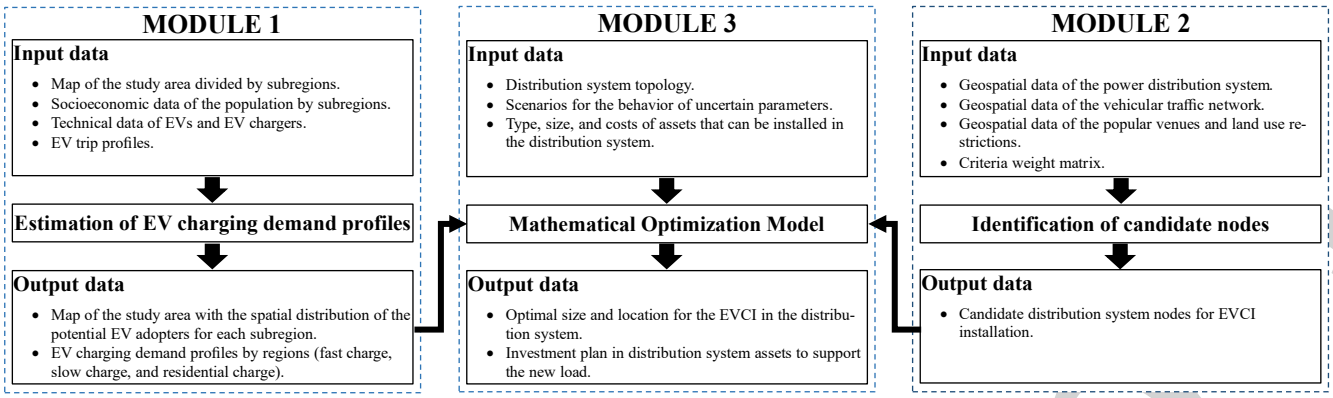

Fig. 1 Flowchart of the proposed framework.

addition to using the overlap and spatial proximity between the nodes of both networks, a multicriteria spatial analysis algorithm is used to determine the candidate nodes for EVCI installation. Furthermore, the multicriteria spatial analysis algorithm allows accounting for preference weighting and the use of land use restrictions to determine where the EVCI should be installed. This task is performed by Module 2.

On the other hand, Module 1 estimates the spatial locations of potential EV users based on socioeconomic data and calculates the EV charging demand profiles. This fact allows for a more adequate sizing of the EVCI.

Therefore, the main contributions of this work are:

1) A new method that uses geospatial data from the distribution system, the road network, popular venues, and land use restrictions is presented to determine candidate nodes to install EVCI.
2) A new framework is proposed to support the integrated planning of distribution systems and EVCI, considering the spatial characteristics of the study area.

This work is an extension of [29] with the following improvements: (i) a more comprehensive literature review of the planning problem; (ii) a summary table of the planning approaches identified in the literature; (iii) the integration of a spatial analysis module to estimate the number of potential EV adopters by subregion; (iv) a trip simulation algorithm to determine EV charging demand profiles by regions, aiding in the optimal sizing of EVCIs; and (v) a broader and more detailed presentation of the results.

C. *Manuscript structure*

The remainder of this paper is organized as follows. First, the proposed framework is presented in Section II. Then, Section III reports numerical results for a Brazilian city. Finally, some relevant conclusions are drawn in Section IV.

II. Proposed Framework

The proposed framework consists of three modules. The first module utilizes socioeconomic data to estimate the number of potential EV adopters by subregions. Following this, a trip simulation algorithm is employed to calculate EV charging demand profiles by region. The second module utilizes spatial data to pinpoint suitable distribution system nodes for EVCI installation. The module integrates findings from the two previous modules along with the distribution system's physical, operational, and topological characteristics to determine the optimal location and size of the EVCI. Additionally, the third module identifies the necessary network assets to meet the new load. Fig. 1 illustrates the flowchart of the proposed framework, while the individual modules are detailed below.

A. *Module 1. Estimation of EV charging demand profiles*

Estimating EV charging demand profiles involves two key stages. The first stage entails identifying potential EV buyers by analyzing the potential of the EV market, broken down by subregions. In the second stage, EV travel within urban areas is simulated, considering user charging preferences, to estimate the EV charging demand profiles by region. Below are the steps involved in each stage.

1) *Spatial distribution of potential EV adopters*

To obtain a more precise estimate of the potential EV adopters, this study proposes partitioning the study area into small subregions known as census sectors. This subregion disaggregation has several advantages, including the ability to make more accurate predictions based on the individual characteristics of each sector's population. Thus, this module employs a geographically weighted regression (GWR) [30], shown in (1), to calculate the potential EV adopters (Y_d) by census tracts while accounting for the socioeconomic data of the local population and the influence of neighboring census tracts. Such influence is known as spatial interactions [31].

$$Y_d = \beta_d^1 X_d^1 + \beta_d^2 X_d^2 + \beta_d^3 X_d^3 + \cdots + \beta_d^k X_d^k + \epsilon_d \qquad (1)$$

In (1), X_d^k is the socioeconomic variable k of the local population in census tract d, ϵ_d is the numerical error, and β_d^k is a sensitivity parameter associated with each socioeconomic variable X_d^k. β_d^k are calculated as explained in [30].

After estimating the number of potential EV adopters within each subregion, the subregions are aggregated into regions. This aggregation considers both the geographical proximity between the subregions and the similarity in the socioeconomic data of the neighboring subregions. This strategic grouping widens the scope for identifying suitable locations for public EVCI installations, particularly in cases where specific subregions lack appropriate land availability for EVCI installation. It is important to note that the total number of potential EV buyers within each region is calculated by summing up the individual values from each subregion.

2) *EV charging demand profiles by region*

Once the number of potential EV adopters in each region has been determined, the module executes an EV travel simulation algorithm. This algorithm is based on real travel profiles conducted by internal combustion vehicles, along with a decision-making algorithm. The latter determines when EVs should be charged and whether they should be charged using

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a fast, slow, or home charger. As a result, three demand profiles by region are generated: one for fast charging, one for slow charging, and one for residential charging. Below is the algorithm for simulating EV travel and making charging decisions for EVs.

Step 1. Input parameters assignment:

- Each EV is assigned with a battery capacity and an initial state of charge (SOC).

Step 2. Daily travel profile assignment:

- A trip profile is assigned for each EV and each day of the year.
- The profile includes the start time of the first trip, the destination region of each trip, the duration of each trip, the distance traveled, and the waiting time for the next trip.
- Days on which the EV will not travel are also considered.

Step 3. Continuous SOC monitoring during travel:

- During each trip, the battery SOC is continually checked.
- If the SOC is greater than a percentage μ_1 of its capacity, the EV will continue the trip, and the SOC will be updated.
- Otherwise, the EV will be recharged.

Step 4. Recharge management:

- When the EV needs to charge, the algorithm determines the charging location according to the destination region and the type of charger to be used (fast, slow, or home).
- The EV charging profile by region is calculated and updated, as well as the SOC of the battery.
- If the EV is plugged into a slow or home charger, it is charged until the battery reaches a percentage μ_2 of its capacity or until the time of the next trip.
- In the case of the EV being plugged into a fast charger, a charging duration of up to a maximum of 60 minutes is randomly chosen. Charging ends when this time is reached or when the battery reaches a percentage μ_2 of its capacity.

Step 5. Repeat the process:

- This process is repeated for every day, month, and EV.

This algorithm guarantees efficient management of the battery capacity of EVs and adequate planning of their trips throughout the year, optimizing charging when necessary to maintain the operational availability of the EVs.

B. Module 2. Identification of candidate nodes

This module identifies suitable locations for installing EVCI by selecting distribution system nodes that intersect with the road network geographically. Additionally, nodes near the road network are also considered candidates. The module then evaluates and ranks these candidate nodes using a multicriteria spatial analysis algorithm that considers a set of criteria, including popular venues and land use restrictions. The algorithm consists of a weighted linear combination and hierarchical analysis method, enabling an effective evaluation of the candidate nodes.

1) Weighted linear combination

The model known as *weighted linear combination*, as described in (2), provides a means to combine n criteria to determine the overall score of candidate nodes for EVCI installation. Each criterion, denoted by k, is assigned a weight w_k, which indicates its relative importance in the evaluation process. These weights must satisfy (3).

$$V(A_i) = \sum_{k=1}^{n} w_k v(a_{ik}) \times \prod_{j=1}^{m} r_j \tag{2}$$

$$\sum_{k=1}^{n} w_k = 1 \tag{3}$$

In (2) and (3), parameter $v(a_{ik})$ denotes the total score of the candidate node i in terms of its geographic characteristics. $v(a_{ik})$ represents the score function of candidate node i with respect to criterion k. The component r_j is used to consider the m land use restrictions used to determine candidate nodes ($r_j = 0$ for land restricted, whereas $r_j = 1$ for land allowed).

2) Hierarchical analysis method

The hierarchical analysis method determines the weights of the n criteria using comparative analysis, in which comparisons are organized in a matrix $[C]$, as shown in (4). The diagonal elements have a value of 1 and $C_{kp} = C_{pk}^{-1}$. The relative weights C_{kp}, which compare the importance of one criterion with respect to another, are determined as described by [32]. The decision-makers provide the comparison matrix $C_{n\times n}$, and the criteria weighting vector e_n can be calculated using the procedure outlined in [32]. To calculate this vector, the Euclidian distance between the spatial location (x_i, y_i) of candidate node i and the spatial location of the criteria used (such as main avenue, shopping mall, office building, residential areas, hospitals, universities, etc.) is computed.

$$v(a_{ik}) = \begin{bmatrix} 1 & C_{12} & \cdots & C_{1n} \\ 1/C_{12} & 1 & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1/C_{1n} & 1/C_{2n} & \cdots & 1 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} \tag{4}$$

3) Steps of the multicriteria spatial analysis algorithm

The algorithm goes through the following steps.

- *Step 1:* Using the spatial superposition of the distribution system and the road network, perform a preliminary selection of candidate nodes for EVCI installation.
- *Step 2:* Define a set of criteria and their weights that will be used to qualify the candidate nodes.
- *Step 3:* Define the matrix $[C]$, calculate the vector of Euclidean distances, and calculate $v(a_{ik})$ using (4).
- *Step 4:* Compute the total score of each candidate node by using (2) and classify them according to the total score.

C. Module 3. Mathematical optimization model

The mathematical optimization model is built upon the one introduced in [33]. Several assumptions form the foundation of this model, including (i) adopting a static planning approach, (ii) accounting for uncertainty by utilizing a set of representative scenarios, (iii) representing the operation of the distribution system with a linearized ac model, (iv) the distribution system must use a radial topology, and (v) considering the presence of distributed generation (DG).

1) Objective function

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TABLE II
ASPECTS CONSIDERED IN THE CASE STUDIES

	Module 1	Module 2	Module 3
Case I	×	×	✓
Case II	×	✓	✓
Case III	✓	✓	✓

The objective function (5) aims to minimize the present value of all expected costs at the end of the planning horizon. This objective function comprises three terms: (i) investment costs in distribution system assets, (ii) operation costs of the distribution system, and (iii) investment costs in EVCI infrastructure. The annual investment costs in distribution system assets, c^{NET}, shown in (6), consists of investments in (i) the replacement of overloaded conductors and (ii) the installation of voltage control equipment such as voltage regulators (VRs) and capacitor banks (CBs). The cost, $\mathcal{I}^{CB}$, is used to highlight that installing CBs at different nodes is more expensive than installing CBs at the same node.

The operation costs of the distribution system, $\mathcal{o}^{NET}$, shown in (7), comprise (i) the cost of the energy purchased from the substations; and the cost of the energy purchased from (ii) dispatchable and (iii) nondispatchable DG units.

The annual investment costs in infrastructure for EVCI, shown in (8), consist of two components: (i) investment costs in the land where the EVCI will be installed and (ii) investment costs in the types of chargers (fast/slow).

$$\text{minimize } c^{NET} + \mathcal{o}^{NET} + c^{CS} \tag{5}$$

$$c^{NET} = \mathcal{Y}^F \left[\sum_{ij \in \Psi^B} \sum_{a \in \Psi^C} \left(\mathcal{C}^C_{\dot{a},a} L_{ij} x^C_{ij,a} + \mathcal{C}^{VR}_a x^{VR}_{ij,a} \right) + \sum_{i \in \Psi^N} \left(\mathcal{I}^{CB} x^{CB}_i + \mathcal{C}^{CB} n^{CB}_i \right) \right] \tag{6}$$

$$\mathcal{o}^{NET} = \left[\frac{(1+\Upsilon)^{\mathcal{Y}^F} - 1}{\Upsilon(1+\Upsilon)^{\mathcal{Y}^F}} \right] \times \sum_{i \in \Psi^N} \sum_{s \in \Psi^S} \Delta^T_s \left(\mathcal{O}^E_{i,s} p^{SS}_{i,s} + \mathcal{O}^G_{i,s} p^G_{i,s} + \sum_{r \in \Psi^{RS}} \mathcal{O}^{RS}_{i,r,s} p^{RS}_{i,r,s} \right) \tag{7}$$

$$c^{CS} = \mathcal{Y}^F \left[\sum_{u \in \Psi^U} \mathcal{I}^{LD}_{i,u} x^{LD}_{i,u} + \sum_{v \in \Psi^{EV}} \mathcal{C}^{EV}_v n^{EV}_{i,v} \right] \tag{8}$$

In (5)–(8) parameter $\mathcal{Y}^F$ is the final-year-of-the-planning-horizon. $\mathcal{C}^C_{\dot{a},a}$ is the cost of substituting a conductor of type $\dot{a}$ by type a. L_{ij} is the length of the branches. $\mathcal{C}^{VR}_a$ is the cost of a VR. $\mathcal{C}^{CB}$ indicates the investment costs of CBs. Υ is the annual interest rate. Δ^T_s represents the duration of a scenario. $\mathcal{O}^E_{i,s}$ is the price of energy at a substation node. $\mathcal{O}^G_{i,s}$ is the operation cost of dispatchable DG units. $\mathcal{O}^{RS}_{i,r,s}$ is the operation cost of nondispatchable DG. $\mathcal{I}^{LD}_{i,u}$ is the cost of the types of land for EVCI. $\mathcal{C}^{EV}_v$ is the investment cost of EVs' chargers.

Besides, the variable $x^C_{ij,a}$ indicates the presence of a conductor of type a on branch ij. $x^{VR}_{ij,a}$ indicates the presence of a VR of type a on branch ij. x^{CB}_i indicates the CB installation at node i. n^{CB}_i indicates the number of CB's modules connected at node i. $p^{SS}_{i,s}$ is the active power from substations. $p^G_{i,s}$ is the active power from dispatchable DG units. $p^{RS}_{i,r,s}$ is the active power from nondispatchable DG units. $x^{LD}_{i,u}$ indicates the purchase of the land of type u, at node i for EVCI. $n^{EV}_{i,v}$ is the number of EVs' chargers of type v installed.

2) Constraints for the distribution system operation

The formulation used to describe the operation of the distribution system relies on expressions (7)–(25) introduced in [33]. These expressions pertain to the active and reactive power balance and voltage drop in the system, obtained through the application of Kirchhoff's laws. Furthermore, these constraints consider physical and operational limits, such as current and voltage limits and substation capacity.

3) Constraints for investment in power system assets

Constraints used to represent investments in distribution system assets are taken from [33], where (17) models the investment in conductors, (18)–(21) model VRs installation, and (22)–(27) represent the installation and operation of CBs.

4) Constraints for the operation of renewable DG units

While this study does not address the sizing and location of renewable distributed generation (DG), it does consider the presence of these technologies. As a result, it is the utility's responsibility to plan the operation of the DG units. For dispatchable renewable DG, the study considers using a set of biomass-based synchronous generators, while for nondispatchable renewable DG, the study considers the presence of photovoltaic (PV) panels and wind turbines (WT).

The operation of DG units is represented by using the formulation provided in [33]. Thus, (26)–(29) are used to represent the operation of biomass generators, while (30)–(33) are used to model the operation of PV panels and WT.

5) Constraints for investment in EVCI

Investing in EVCI involves both the purchase of land and chargers. The cost of the land varies depending on its size and location, while the number of chargers that can be installed on each land is limited based on their size. The formulation for representing the investment in EVCI is based on the expressions (77)–(85) presented in [33].

6) Characterization of the uncertainties

This study examines five uncertain parameters: electricity demand, wind speed, solar irradiation, energy prices, and EVCIs' load profiles in each region. These parameters' behavior is analyzed using a stochastic scenario-based programming framework. Historical data forms the basis for scenarios, which are categorized into four groups corresponding to the seasons (spring, summer, fall, and winter), each further divided into day and night sub-groups. To segment the data into *k clusters* within each sub-group, the *k-means* algorithm [34] is applied. Consequently, a set of $(4 \times 2 \times k)$ scenarios depict the annual variation in the uncertain parameters.

III. TESTS AND RESULTS

The proposed framework's applicability and effectiveness were evaluated using socioeconomic and geospatial data from a medium-sized Brazilian city with a population of around 230,000. The urban area of this city is divided into approximately 300 subregions. Socioeconomic characteristics were drawn from the population census conducted by the Brazilian Institute of Geography and Statistics [35]. Furthermore, the city's power distribution system comprises four substations and 28 medium-voltage feeders.

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Modules 1 and 2 were implemented using ArcGIS 10.5 software from ESRI [36]. The information was processed with the help of the ArcPy tool and the Python programming language. Additionally, the mathematical model of Module 3 was written in the AMPL [37] and solved using CPLEX v20.1 [38]. Simulations were performed on a computer with a 3.4 GHz AMD Ryzen™ 9 5950X processor and 32 GB of RAM.

This article presents the findings from three case studies, summarized in Table II, on the integrated planning of distribution systems and EVCI. In Case I, Modules 1 and 2 are not considered for planning. In Case II, Module 1 is not considered. Finally, in Case III, the three Modules were integrated.

The three case studies consider the same number of EVs adopted and the same EV charging demand profiles. However, they present the following differences:

In Case I, all nodes of the distribution system are considered candidates for EVCI installation. In addition, the optimization model sizes the EVCI in each node in the most economical way for the system. In Case II, the candidate nodes for installing EVCI are restricted to those with a better score in terms of their spatial location. Even so, the optimization model also sizes the EVCI without restrictions. In Case III, the candidate nodes for installing EVCI are also restricted to the best spatially located ones. Furthermore, the sizing of EVCIs must be done considering the spatial location and the number of potential EV adopters in each subregion.

A. Input data

The input data for Module 1 includes the following information. To estimate the number of potential EV buyers by subregions, four socioeconomic variables were analyzed: (i) the household head's monthly income, (ii) the household head's educational level, (iii) the number of cars per household, and (iv) the population's interest in innovative technological devices. Ref. [39] contains the spatial database with the shapefile files of the study area, with their respective division by regions and subregions and the spatial distribution of the socioeconomic variables of the population. Furthermore, three types of EVs with battery capacities of 35, 40, and 50 kWh were considered to estimate the EV charging demands. Additionally, a nominal power of 50 kW was accounted for fast chargers, while a nominal power of 10 kW was taken into consideration for slow and residential chargers. The travel profiles of the EVs were sourced from [40]. In these travel profiles, we know the number of trips per day, the destination of each trip, the start time of the first trip of the day, the duration of each trip, and the waiting time to make the next trip.

The input data for Module 2 includes geospatial data of the power system, the road network, population venues, and protected areas. Fig. 2 presents a map of the city under study with geospatial data for the power system and the road network. Additionally, the distribution system nodes have been classified according to five criteria, which consider the nodes' proximity to (i) main highways, (ii) office buildings, (iii) shopping centers, (iv) residential areas, and (v) universities and hospitals. Table III summarizes the matrix representing the relative importance of each criterion in relation to the others. The land use restrictions applied to declassify nodes consider the proximity to (i) protected areas and forests, (ii) national parks and cultural heritage sites, (iii) waterways and streams, and (iv) cultivated areas and geologically unstable areas. Fig. 3 displays a city map indicating the geographical location of popular venues as well as the protected areas that will be considered for land use restrictions. Ref. [39] contains the spatial database with the shapefile files of the study area, with the geospatial location of each of the popular venues, as well as the spatial locations of each of the restricted areas. The GIS utilizes layers to integrate geospatial data for each criterion, encompassing both popular venues and land use restrictions.

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The input data for Module 3 includes the following parameters. The results are shown for a single medium-voltage feeder that provides electricity to residential consumers. The simulations for the distribution system consider maximum and minimum voltage magnitudes of 1.05 p.u. and 0.95 p.u., respectively. Ref. [39] encompasses all the technical data of the distribution system utilized in the simulations. Furthermore, this document provides both technical and financial data concerning the assets that can be installed.

B. Results

Fig. 4 shows the results of the potential EV adopters. The results indicate that the adoption of EVs varies significantly across different subregions, attributable to the diverse socio-economic features of the local population. This information is critical for appropriately scaling the EVCI according to the number of chargers necessary to meet the demand of EV adopters in each region.

The candidate nodes for EVCI installation, calculated by Module 2, are divided into four groups based on their score: (i) unsuitable, (ii) lowly suitable, (iii) suitable, and (iv) highly suitable. In this work, only the nodes classified as suitable and highly suitable are used as input data for Module 3.

Table IV presents the investment costs for the three case studies considered, while Table V shows the investment plan in network assets for each case. In addition, for each case study, Figures 5, 6, and 7 display the results of the distribution system nodes where the EVCI will be installed.

C. Discussion

Several aspects can be observed when comparing the results of Cases I and II. To begin with, while the cost of network planning is lower in Case I than in Case II, the results of Case I in terms of EVCI location may be unfeasible. This is because the optimization model does not consider the geographical characteristics of the study area, which may result in recommendations for installing the EVCI in subregions with land use restrictions or very far from popular venues, such as main highways, office buildings, and shopping malls.

The results, however, show that even if the same number of fast and slow chargers are installed in both cases, the network planning cost varies. The network planning cost in Case II is 3.51% higher than in Case I, indicating that the location of EVCI has a different impact on network planning. Furthermore, the cost of land for the installation of EVCI varies depending on the location because different costs for land are considered depending on the location.

Although the planning costs for Case I are the lowest, fig. 5 reveals several important observations about the results obtained. Firstly, EVCI 3 and 4 were installed in protected areas, rendering this solution unfeasible in real life. Furthermore, EVCI 2 and 5 were located far from the city's popular avenues, thereby complicating access for EV users via the road network. These findings underscore the importance of employing Module 2 of the proposed framework for the proper location of EVCI, considering both proximity to popular avenues and land use restrictions. Additionally, fig. 5 indicates that EVCI 2, 5, and 6 were installed in subregions without potential EV adopters, rendering the solution inefficient. Ideally, EVCI should be situated in subregions with a higher number of EV adopters.

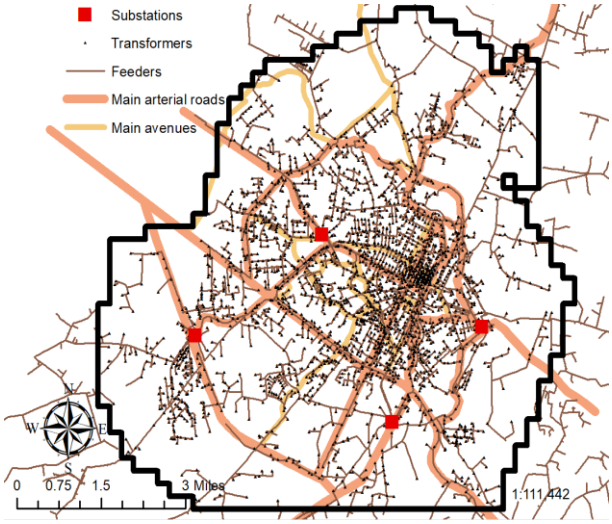

Fig. 2 Geospatial data of the power system and the road network

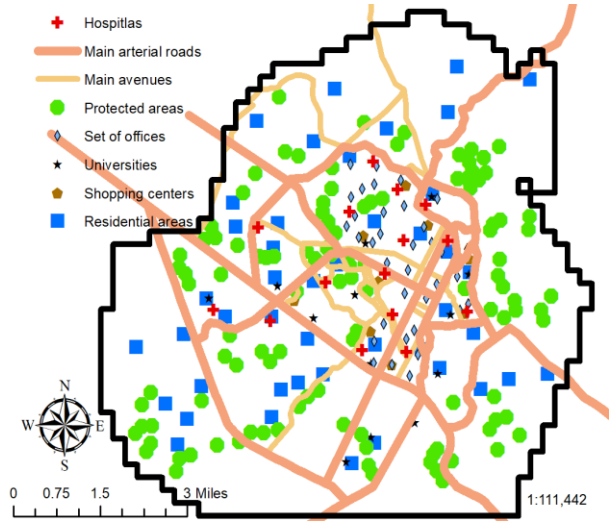

Fig. 3. Geospatial location of popular venues and land use restrictions.

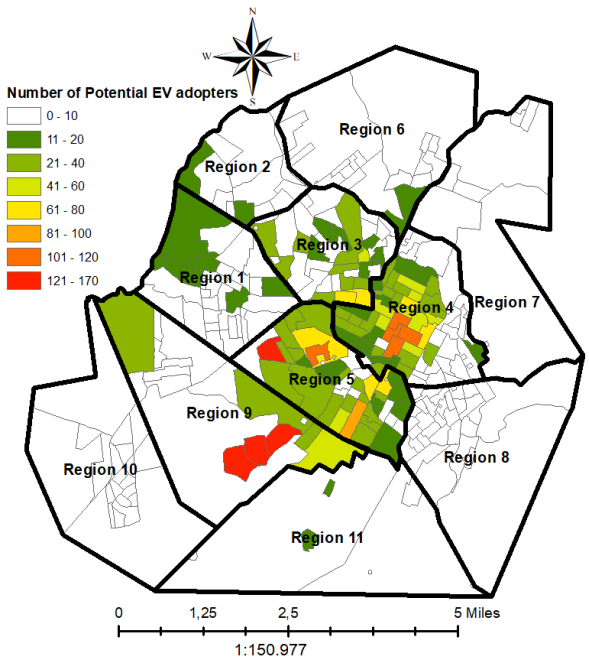

Fig. 4 Spatial distribution of potential EV adopters.

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TABLE III
CRITERIA, WEIGHTS, AND RECIPROCAL MATRIX

Criteria	Weights (%)		1	2	3	4	5
Main highway	30	1	1	7	2	3	5
Office building	25	2	1/7	1	2	1/3	6
Shopping malls	15	3	1/2	1/2	1	2	3
Residential areas	20	4	1/3	3	1/2	1	8
Universities	10	5	1/5	1/6	1/3	1/8	1

TABLE IV
OVERVIEW OF THE SOLUTIONS

Network planning results		Case I	Case II	Case III
Investment cost (10^3 USD)	Conductors	115.98	156.40	145.28
	VRs	99.15	104.14	172.93
	CBs	103.35	151.42	150.02
Total investment cost (10^3 USD)		318.48	411.96	468.23
Total operation cost (10^3 USD)		31,852.90	32,889.83	36,916.40
Total cost (10^3 USD)		32,171.38	33,301.79	37,384.63
EVCI planning results		**Case I**	**Case II**	**Case III**
Investment cost (10^3 USD)	Fast chargers	371.64	371.64	395.59
	Slow chargers	491.43	491.43	542.83
	EVCI installation	1,379.31	1,516.58	2,118.05
Total investment cost (10^3 USD)		2,242.38	2,379.65	3,056.47

Some other phenomena can be observed when comparing the results of Cases II and III. First, even though both cases considered the same number of potential EV buyers, the results of Case III installed more slow and fast chargers than Case II. This is because the optimization model recommends installing at least one EVCI in each subregion of the study area. It should be noted that the number of chargers installed in each EVCI is proportional to the number of EV adopters in each region. It is important to note that the optimization model recommends placing more chargers in EVCI that serve regions with a higher concentration of potential EV adopters.

Although Fig. 6 shows that, for Case II, the EVCI were installed in strategic locations, such as near popular avenues and in compliance with protected areas and other land use restrictions, it also reveals that EVCI 2, 6, and 8 were placed in subregions without potential EV adopters. Furthermore, EVCI 3 was sized larger than EVCI 7, knowing that EVCI 7 serves a subregion where there is a greater number of EV adopters. This situation renders the solution inefficient. These findings underscore the importance of employing Module 1 of the proposed framework for the proper sizing of EVCI, considering the spatial location of potential EV adopters.

Finally, although Case III is the most expensive of all, economically, Figure 7 shows that both the location and sizing of the EVCI represent the best solution among the three study cases when the geographical and spatial characteristics of the area of study are considered.

IV. CONCLUSIONS

This article presented an innovative framework for planning electric vehicle charging infrastructure (EVCI) and distribution systems through GIS-based spatial analysis. The framework is divided into three modules. The first module employs socioeconomic data to estimate the spatial distribution of potential EV users and utilizes a travel simulation algorithm to generate charging demand profiles for these vehicles. The second module analyzes spatial data from the study area to identify the most suitable distribution system nodes for EVCI installation. The third module integrates the findings from the previous two modules with the physical, operational, and topological characteristics of the distribution system to determine the optimal locations and sizes for EVCI. Furthermore, this module identifies the network assets necessary to meet the new EV charging demand.

The proposed framework was tested using socioeconomic and spatial data from a medium-sized city in Brazil. The findings suggest that integrating the spatial characteristics of the study area can lead to more effective solutions for the location and size of EVCI. However, it was also observed that planning EVCI with these spatial aspects in mind might result in

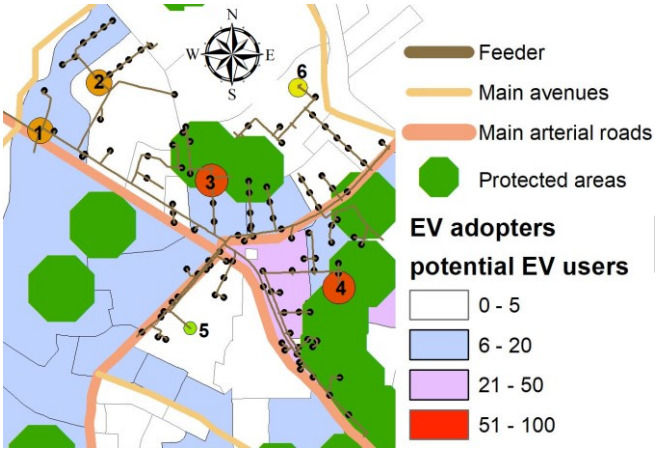

Fig. 5 Results for Case I.

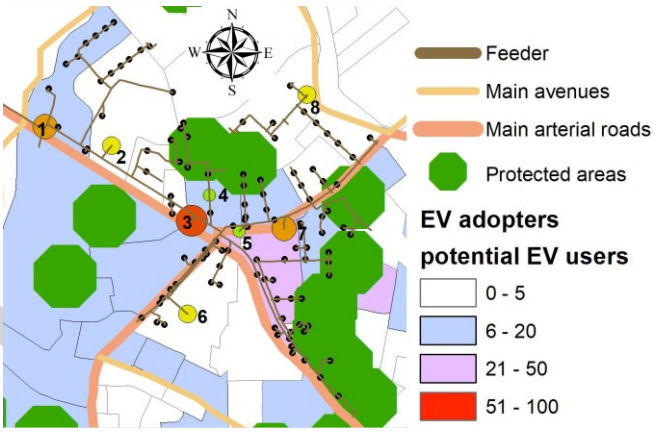

Fig. 6 Results for Case II.

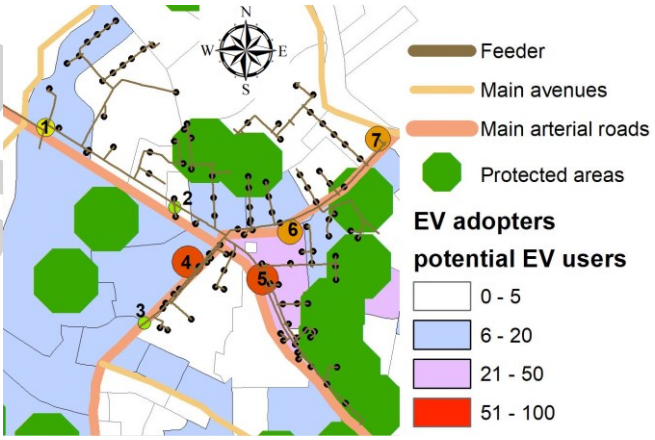

Fig. 7 Results for Case III.

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TABLE V
INVESTMENTS PLANS

Investment option	Case I	Case II	Case III
Conductors installed (branch ij, type)	01-02, III; 02-03, II; 03-04, II.	01-02, III; 02-03, III; 03-04, III; 04-05, III; 05-06, II.	01-02, III; 02-03, III; 03-04, III; 04-05, II.
VRs installed (branch ij, type)	07-08, II; 14-15, I.	07-08, II; 14-15, I; 23-24, I; 37-38, II.	07-08, III; 14-15, I; 23-24, I.
CBs installed (node i, quantities)	25, 5; 30, 2; 41, 4; 55, 5; 79, 6.	09, 5; 25, 3; 36, 5; 41, 6; 65, 6; 79, 7.	09, 5; 25, 3; 36, 5; 41, 6; 65, 6; 79, 7.
Fast EV charger (node i, quantities)	10, 8; 21, 14; 33, 15; 45, 5; 64, 10; 79, 4; 88, 9.	12, 8; 21, 14; 35, 15; 48, 5; 64, 10; 80, 4; 89, 9.	10, 8; 22, 10; 33, 15; 10, 12; 22, 14; 31, 10; 45, 10; 62, 5; 71, 9; 88, 9.
Slow EV charger (node i, quantities)	10, 6; 21, 10; 33, 12; 45, 15; 64, 12; 79, 6; 88, 7.	12, 6; 21, 10; 35, 12; 48, 15; 64, 12; 80, 6; 89, 7.	10, 6; 22, 9; 33, 12; 10, 11; 22, 11; 31, 10; 45, 4; 62, 12; 71, 6; 88, 6.

higher economic costs than planning without such considerations. It is important to note that this will not always be the case, as there may be instances where the optimal locations of distribution system nodes coincide with the best geographical locations for EVCI installation.

One of the main advantages of the proposed framework is that the optimization model retains the autonomy to determine the optimal size and location of EVCI to minimize the investment and operational costs of the distribution system. The spatial analysis algorithm identifies the EV charging demand profiles by region and selects the best candidate system nodes for EVCI installation based on a set of criteria, including popular venues, land use constraints, and the spatial distribution of EV adopters.

For future research, it is proposed to extend the model towards a multistage planning approach, aiming to determine the optimal timing for installing network assets and EVCI. It is also suggested to develop a method that includes consideration of EVCI not owned by the utility to identify the complexities of joint planning involving independent owners.

Dynamic thermal classification (DTC) is an innovative solution to optimize the power grid by adjusting distribution line capacities in real time based on environmental conditions. Factors such as temperature, wind speed, and solar radiation directly affect the thermal dissipation of lines, allowing their operating limits to be adapted to the specific circumstances of the moment. This approach significantly improves operational efficiency by overcoming the limitations of static thermal classifications, which use conservative values to ensure safety in any scenario. According to studies, DTC can increase transmission capacity by 10% to 40% under favorable conditions [41], [42], reducing operational constraints and postponing costly infrastructure investments.

Although our study used static thermal classifications to simplify mathematical modeling, we recognize the benefits of incorporating DTC. This integration would allow for a more accurate representation of network capacities, especially in contexts of climate variability. Therefore, we suggest that future research combines CTD with our proposed framework, which will not only broaden the applicability of the results but also contextualize our findings within current developments in the electricity sector.

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